

PHYS400
SPECIAL PROBLEMS IN PHYSICS
PROJECT PROPOSAL

Project Title	Variational Approach to the Energy Spectra of Heavy Mesons
Student Name	Levent Kaan Oğuz
Student ID	2555365
Advisor	Altuğ Özpineci
Field of study	Theoretical Particle Physics
Type of study	Theoretical/Computational
Submission Date	March 27, 2026

Abstract

Heavy mesons are mesons that contain either one of the charm, bottom or top quarks and one antiquark version of the three. In this project, we will investigate the bound-state energies of heavy quarks using non-relativistic potential quark models. While Quantum Chromodynamics explains the strong interaction, for low energy scale, solving it is quite complex. We will use an effective potential so called Cornell potential to approximate quark interactions within non-relativistic Schrödinger Equation. Our main work will govern application of Ritz variational method to solve for heavy mesons to find their energy eigenvalues. We will try various trial functions such as Gaussian, exponential, their superpositions and minimize our change in expectation of energy by our free parameters, imposing a gradient-descent-like method. There has been other We will be using Python and SciPy module to computationally automate the minimization, calculating energy eigenstates numerically and comparing our results with others. Expected outcome is finding energy levels that aligns well with empirical results and published results such as Particle Data Group tables. Also we will be testing the methods limits and whether or not it is more feasible than other methods.

Introduction

The dynamics of hadronic states are described by Quantum Chromodynamics (QCD), at high energies this theory is characterized by asymptotic freedom and color confinement at macroscopic scales. In the low-energy scale, strong coupling constant, α_s is no more a constant but it also diverges, causing perturbative methods to fail and creates necessity for the lattice QCD simulations or phenomenological field models Elpe (2016).

Mesons are particles that contain quark-antiquark pair, which can be stated as $q\bar{q}$. By heavy mesons, we mean that mesons containing one or two of the charm or bottom quarks, these quarks, due to heavy quark symmetry assures us that constituent velocities are low enough to simplify our problem to non-relativistic scale Griffiths (1987). An example can be given as J/ψ , which is an 1^3S_1 state of $c\bar{c}$. Since such mesons are heavy, we can use time-independent Schrödinger equation with a confining potential at long distances and asymptotic freedom in short distances Griffiths (1987). Cornell potential, $V(r) = \frac{-4}{3}\alpha_s/r + \sigma r + \text{constant}$, captures both very well Eichten et al. (1975). Due to

how this potential is stated, exact solution is impossible. This creates the necessity for approximate methods such as perturbation, or in this case, Ritz variational method. In Cornell potential, Coulomb-like term arises from single-gluon exchange, this term dominates the effective potential in short distances. The linear term, however, which is scaled by QCD string tension parameter, dominates effective potential at long distances Godfrey and Isgur (1985). Constant parameter is added due to constituent quark masses. When we use only this potential without any fine or hyperfine corrections, we assume that such affects are minimal Griffiths (1987). However, while Cornell potential helps us to identify the behaviour of heavy quarkonium, our calculations are expected to deviate from empirical values if we do not implement $O(\alpha_s^2)$ corrections due to relativistic adjustments and fine and hyperfine corrections Napsuciale et al. (2025).

Using Ritz variational method, there has been successful trials using spherical quantum harmonic oscillator eigenstates that includes Laguarre polynomials Akbar et al. (2025) and power or logarithmic fits Mutuk (2018). In this project, we will try more trial functions such as exponentials and Gaussians, or try Gaussian Expansion Method Hiyama et al. (2003). After all that, we will calculate our errors with experiments using the Particle Data Group data and compare our results with other sources to identify the feasibility of this method.

In this project we will create a pipeline-like tool to apply trial functions through Ritz variational method to automatize the trial of functions and finding optimum state functions. After that we will be adding fine-structure corrections to such states and minimizing χ^2 error referenced to Particle Data Group data Navas et al. (2024). Once this seemed to be done, we can continue to expand our view to multi-quark states and it may also help for the solutions to exotic states such as X(3872).

Research Question

Is Ritz variational method with $O(\alpha_s^2)$ corrections usable to predict heavy meson energy spectra using a wide range of trial wavefunctions better than the perturbative or supersymmetric approaches?

This project will extend the scope of computationally tried trial functions function and find exactly how much discrepancy will be reduced after relativistic and fine structure corrections.

Method	E_0	E_1	E_0 Absolute Error	E_1 Absolute Error
Perturbation (1st)	75.50000	376.50000	7.24e+01	3.65e+02
Perturbation (2nd)	-26174.50000	-205873.50000	2.62e+04	2.06e+05
Variational	3.19244	11.32488	6.11e-02	1.38e-01
Shooting	3.13138	11.18725	3.25e-06	2.33e-05
Matrix Finite Differences	3.13138	11.18723	-	-

Table 1: Quantum harmonic oscillator with quartic disturbance calculations using $\hbar = \omega = m = 1$ and $\sigma = 100$. Calculations are done with variational method, perturbation theory up to first and second-order corrections, shooting method and matrix finite differences method. Errors are referenced to finite differences method.

Methodology

To find energy levels of heavy mesons, we will use Cornell Potential with reduction to one body problem that gives the Hamiltonian

$$H = \frac{p^2}{2\mu} - \frac{4}{3} \frac{\alpha_s}{r} + \sigma r + \text{constant} \quad (1)$$

where p is the relative momentum, μ is the reduced mass, α_s is the strong coupling constant, σ is the QCD string tension constant, r is the relative distance A Alkathiri et al. (2024).

Using Ritz variational method, we will be using trial wavefunctions, such as Gaussian or exponential or any superpositions of such functions, with one or more free parameters to find expectation of the Hamiltonian, $\langle H \rangle$, which is also equal or greater than the ground-state energy, leading to Equation 2 Sakurai and Napolitano (2020).

$$\langle H \rangle = \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_{\text{ground}} \quad (2)$$

Then, we will use a gradient-descent-like method to optimize for the free parameters, β_i , using the knowledge we must have an exact energy level and changes in energy level with respect to this parameter is a cost function. We basically solve Equation 3. After that, substitute such α_i back to Equation 1.

$$\frac{\partial \langle H \rangle}{\partial \beta_i} = 0 \quad (3)$$

For observational purposes, we can apply this method to quantum harmonic oscillator problem with quartic disturbance, i.e. $H = \frac{p^2}{2m} + \frac{1}{2}mw^2x^2 + \sigma x^4$, with Gaussian trial function, $N\exp(-\beta x^2)$. One can observe the results are promising from Table 1 and Figure 1 even though perturbative method stays behind due to how disturbance is noticeable.

Applying this method, we can find an upper bound for the energy levels of non-relativistic

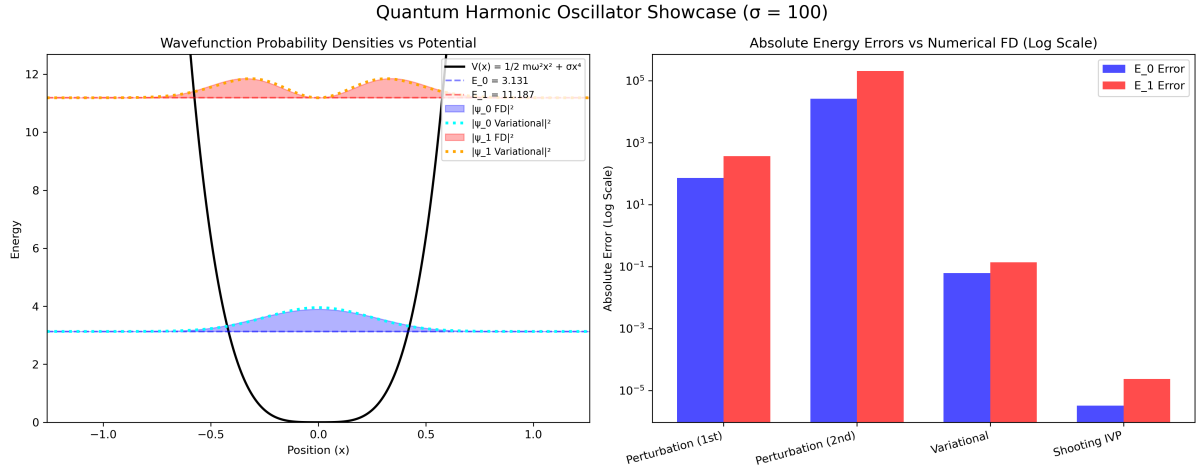


Figure 1: Graphical showcase for the calculated wave functions and energies with errors.

hadrons, such as charmonium and bottomonium. Once the unperturbed parameters are set, we will try to implement $O(\alpha_s^2)$ corrections and fine and hyperfine-structure interactions to split degenerate states such as $\chi_{c0}, \chi_{c1}, \chi_{c2}$ Napsuciale et al. (2025).

To do all this, Python will be vastly used with SciPy module. This package will help us optimize our parameters using `scipy.optimize.minimize` and, even if we will not be using these methods, solve Cornell Potential by shooting using `scipy.integrate.solve_ivp` or finite differences using `scipy.linalg.eigh_tridiagonal`.

After all done, we will compare this results with others that used other methods like perturbation or other trial wavefunctions or with the experiment results. For example, we will compare mass of η_c (0^+0^{-+}), which is $2984.1 \text{ MeV} \pm 0.4 \text{ MeV}$ Navas et al. (2024), or J/ψ , (0^-1^{--}), which has mass of $3096.900 \pm 0.006 \text{ MeV}$ Navas et al. (2024), with our results.

While comparing our results with empirical counterparts, we will initiate a χ^2 minimization algorithm to optimize the free parameters of the Hamiltonian, i.e. quark masses, strong coupling, QCD string tension, zero-point correction. The algorithm uses the Equation 4, where $M_{\text{exp}, i}$ is the experimental targets of masses and $\sigma_{\text{exp}, i}$ represents the uncertainties from Particle Data Group tables Navas et al. (2024). The process will be iteratively applying gradient-descent to χ^2 to find optimal parameter set to minimize the variance and align our model with empirical results Bevington and Robinson (2002).

$$\chi^2 = \sum_i \frac{(M_{\text{theory}, i} - M_{\text{exp}, i})^2}{\sigma_{\text{exp}, i}^2} \quad (4)$$

Expected Results

We expect variational method to be more successful on predicting energy levels due to how perturbative methods require other terms to be significantly small compared the *dominant* potential term. This method provides us a general result for the state functions and this can simply be used for finding energy levels without requiring for *other* terms to be significantly small.

After we find calculated eigenvalues we will add constituent quark masses to them and the results are expected to be align with the experimental data shared by Particle Data Group. Any discrepancies should suggest that there must be corrections to the non-relativistic approximation and if we manage to implement $O(\alpha_s^2)$ corrections, we expect such discrepancies to be neglectable.

After doing this project, we plan to generalize our problem to multi-quark systems such as baryons or tetraquarks.

Timeline

		Semester Plan (15 Weeks)														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Literature Review																
Computational Setup & Trial Function Optimization																
Fine-Structure Integration and Benchmarking																
Writing & Poster Preparation																

Work Packages

WP1 - Literature Review

We will try to understand fundamental theory and computational tools for the project. We will try to apply Ritz variational method to a toy problem, e.g quantum harmonic oscillator with quartic disturbance, and compare it with perturbative method and find how much it differs from finite difference integration. We will start using mentioned `scipy.optimize.minimize` here throughout.

We will be doing theoretical review from several papers, containing but not limited to Ak-

bar et al. (2025), Godfrey and Isgur (1985), Napsuciale et al. (2025), Eichten et al. (1975).

WP2 - Computational Setup & Trial Function Optimization

After doing our toy problem, we will try to automatize our procedure by a pipeline-like modular structure that takes corresponding trial wavefunctions from predefined Python functions and calculates optimized parameters and energy eigenstates. Main objective is to test a variety of wave functions such as Gaussian, exponential, their superposition, and also to understand what are the limits of variational method through this problem.

WP3 - Fine-Structure Integration and Benchmarking

We will rigorously derive $O(\alpha_s^2)$ and fine and hyperfine-structure corrections such as spin-spin and spin-orbit couplings. We will then add these corrections to the results from WP2 and numerically find exact expectations of energy levels. After doing this we expect to explicitly differentiate and calculate energy splittings of vector and pseudoscalar states and fine-structure triplets of P-wave states.

After these results are taken, we will compare this with latest Particle Data Group Navas et al. (2024) tables for the $c\bar{c}$ and $b\bar{b}$ families with different angular momenta. We will try to use χ^2 minimization to optimize our Hamiltonian parameters, such as masses, strong coupling constant, and QCD string tension. We also plan to check where non-relativistic approximation fails so that we may also be able to identify how much we should account for relativistic corrections for heavy-light mesons.

As a final work, we will try to lay out a further progress explanation to how to generalize this problem to baryons, three-body quark systems.

References

- A Alkathiri, A., El-Sherif, L. S., Bakry, A. M., Ismail, A., and Allosh, M. (2024). Puzzle of the cornell potential on the heavy and heavy-light meson spectra. *Physica Scripta*, 99(10):105305.
- Akbar, N., Shafaq, B., Zahra, S., and Mir, A. (2025). Mass spectrum, root-mean-square radii, form factors, and charge radii of mesons. *Journal of the Korean Physical Society*, 86(11):1037–1047.

- Bevington, P. R. and Robinson, D. K. (2002). *Data reduction and error analysis for the physical sciences*. McGraw-Hill Professional, New York, NY, 3 edition.
- Eichten, E., Gottfried, K., Kinoshita, T., Kogut, J., Lane, K. D., and Yan, T. M. (1975). Spectrum of charmed quark-antiquark bound states. *Physical Review Letters*, 34(6):369–372.
- Elpe, A. (2016). Heavy mesons in heavy quark effective theory. Master’s thesis, Middle East Technical University.
- Godfrey, S. and Isgur, N. (1985). Mesons in a relativized quark model with chromodynamics. *Physical Review D*, 32(1):189–231.
- Griffiths, D. (1987). *Introduction to Elementary Particles*. Wiley.
- Hiyama, E., Kino, Y., and Kamimura, M. (2003). Gaussian expansion method for few-body systems. *Progress in Particle and Nuclear Physics*, 51(1):223–307.
- Mutuk, H. (2018). S-wave heavy quarkonium spectra: Mass, decays, and transitions. *Advances in High Energy Physics*, 2018:1–12.
- Napsuciale, M., Rodríguez, S., and Villanueva-Gutiérrez, A. (2025). Complete analytical solution to the cornell potential and heavy quarkonium structure. *Physical Review D*, 111(11).
- Navas, S. et al. (2024). Review of particle physics. *Phys. Rev. D*, 110(3):030001.
- Sakurai, J. J. and Napolitano, J. (2020). *Modern Quantum Mechanics*. Cambridge University Press.